

Median Index Preservation under Bounded Score Perturbations

Research Architecture Runtime
operator/median

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Abstract

The median index is the unique strict k -th order statistic for the conventional median rank k . When every pairwise score gap exceeds 2ε , that index is invariant under every additive perturbation with $\|\delta\|_\infty \leq \varepsilon$. The claim is Lean-certified (`LEAN_FULL`) as a definitional reduction of the k -th order-statistic margin theorem.

This theorem is: Reduction.

1 Problem

The median operator on a finite score vector $s \in \mathbb{R}^n$ ($n \geq 2$) returns the unique index i^* that is the strict k -th smallest coordinate of s , where k is the conventional median rank (so that exactly k scores are strictly smaller than s_{i^*} , and no other index shares the value s_{i^*}). The selected object is this index (equivalently, the unique median position under a strict-order presentation), not a set-valued median band.

2 Stability notion

Admissible perturbations are additive score noise $\delta \in \mathbb{R}^n$ in the closed ℓ_∞ ball $\|\delta\|_\infty \leq \varepsilon$ for a fixed $\varepsilon \geq 0$. Stability means: if i^* is the unique strict k -th smallest index of s and every pairwise gap of s exceeds 2ε , then i^* remains the unique strict k -th smallest index of $s + \delta$ for every such δ . Sharpness: if some rival index lies within distance 2ε of s_{i^*} , an admissible δ destroys uniqueness of the strict k -th position.

3 Definitions

Definition 1 (Strict k -th smallest). *Fix $n \geq 2$, $s : \{0, \dots, n-1\} \rightarrow \mathbb{R}$, and $k < n$. An index i is the strict k -th smallest if $\#\{j : s_j < s_i\} = k$ and $s_j = s_i$ forces $j = i$.*

Definition 2 (All-gaps margin). *Scores satisfy $\text{AllGapsExceed}(s, g)$ if $|s_p - s_q| > g$ whenever $p \neq q$.*

Definition 3 (ℓ_∞ ball). *$\|\delta\|_\infty \leq \varepsilon$ means $|\delta_t| \leq \varepsilon$ for every coordinate t .*

Assumptions. $n \geq 2$, $\varepsilon \geq 0$, a unique strict k -th smallest index exists, and (for invariance) $\text{AllGapsExceed}(s, 2\varepsilon)$.

4 Theorem

Theorem 4 (Median / k -th margin invariance). *Let $n \geq 2$, $\varepsilon \geq 0$, and let i be the strict k -th smallest index of s . If $\text{AllGapsExceed}(s, 2\varepsilon)$, then for every δ with $\|\delta\|_\infty \leq \varepsilon$, the same index i is the strict k -th smallest index of $s + \delta$.*

Theorem 5 (Sharpness). *Under the same standing hypotheses on (n, s, k, i) and $\varepsilon \geq 0$, if there exists $j \neq i$ with $|s_i - s_j| \leq 2\varepsilon$, then there exists δ with $\|\delta\|_\infty \leq \varepsilon$ such that i is not the strict k -th smallest index of $s + \delta$.*

5 Intuition

An ℓ_∞ adversary can decrease one coordinate by at most ε and increase another by at most ε , consuming at most 2ε of any pairwise gap. If every gap is strictly larger than 2ε , no pairwise order can reverse and no tie can form, so the count of scores below s_i and the uniqueness of s_i are preserved. If some gap is at most 2ε , a midpoint collision forces a tie and uniqueness fails.

6 Examples

Example 6. Take $n = 3$ scores $(s_0, s_1, s_2) = (1, 4, 7)$ and median rank $k = 1$ (unique middle value at index 1). Pairwise gaps are 3, 6, and 3, so $\text{AllGapsExceed}(s, 2\varepsilon)$ holds whenever $2\varepsilon < 3$, e.g. $\varepsilon = 1$. For any $\|\delta\|_\infty \leq 1$, the middle index remains 1: the perturbed values stay strictly ordered. If instead $\varepsilon = 2$ so that $2\varepsilon = 4$ meets the gap $|4 - 1| = 3 \leq 4$, the midpoint adversary between indices 0 and 1 produces a tie and destroys uniqueness of the strict first order statistic.

7 Proof

Write $s' = s + \delta$ and assume $\|\delta\|_\infty \leq \varepsilon$.

Pairwise order preservation. Fix distinct indices p, q with $|s_p - s_q| > 2\varepsilon$. Without loss of generality suppose $s_p < s_q$, so $s_q - s_p > 2\varepsilon$. Then

$$s'_p - s'_q = (s_p - s_q) + (\delta_p - \delta_q) \leq -(s_q - s_p) + 2\varepsilon < 0,$$

because $|\delta_p - \delta_q| \leq 2\varepsilon$. Hence $s'_p < s'_q$. The symmetric argument gives the converse direction: $s'_p < s'_q$ forces $s_p < s_q$. Thus

$$s_p < s_q \iff s'_p < s'_q.$$

Count preservation. Applying the pairwise equivalence at $q = i$ yields $s_j < s_i \iff s'_j < s'_i$ for every j . Therefore

$$\#\{j : s_j < s_i\} = \#\{j : s'_j < s'_i\}.$$

If i is strict k -th smallest for s , the left-hand side equals k , so the right-hand side equals k .

Uniqueness preservation. Suppose $s'_j = s'_i$. If $j \neq i$, then $|s_j - s_i| > 2\varepsilon$ by the all-gaps hypothesis, and the same ℓ_∞ estimate used above yields $s'_j \neq s'_i$, a contradiction. Hence $j = i$.

Combining the two paragraphs shows i remains the strict k -th smallest index of s' , which is Theorem 4. (Median specialization: take the conventional median rank k ; the argument never uses any property of k beyond $k < n$.)

Sharpness. Let $j \neq i$ satisfy $|s_i - s_j| \leq 2\varepsilon$. Define the midpoint collision

$$\delta_t = \begin{cases} -(s_i - s_j)/2 & t = i, \\ (s_i - s_j)/2 & t = j, \\ 0 & \text{otherwise.} \end{cases}$$

Then $|\delta_t| \leq \varepsilon$ for all t (since $|s_i - s_j|/2 \leq \varepsilon$), and $s'_i = s'_j$. Uniqueness of a strict k -th value at i therefore fails, proving Theorem 5.

8 Formal statement

Lean module: `Research.Operators.Median.Preservation`. Source file: `lean/Research/Operators/Median/Preservation`.
Theorems: `median_margin_invariance`, `median_margin_sharpness`. These are definitional aliases of `KthMarginInvarianceProp` / `KthMarginSharpnessProp` in `Research.Operators.OrderStat.KthMargin`.
Certificate: `lean/certificates/median/median-margin`. Status: `LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).

9 Proof dependencies

- Reduces to `Research.Operators.OrderStat.KthMargin` (`kth_margin_invariance`, `kth_margin_sharpness`).
- Uses the ℓ_∞ gap estimate $|(\delta_p - \delta_q)| \leq 2\varepsilon$ (triangle inequality for absolute values).
- Uses the definitions of strict k -th smallest, all-gaps margin, and midpoint collision adversary from that core.
- No other operator theorems are required.

10 Consequences

Median is the canonical specialization of unique strict order-statistic selection. The same margin theorem governs `quantile`, `percentile`, and `kth-order-statistic` (all reduce to the same k -th core). Ranking and top- k operators use the related pairwise ranking core rather than this count-based k -th statement; together they form the order-statistic layer of the primitive library. Any pipeline that selects a median index under bounded score noise may invoke Theorem 4 as the stability certificate.